Exploring Pretraining via Active Forgetting for Improving Cross Lingual Transfer for Decoder Language Models

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Abstract

Large Language Models (LLMs) demonstrate exceptional capabilities in a multitude of NLP tasks. However, the efficacy of such models to languages other than English is often limited. Prior works have shown that encoder-only models such as BERT or XLM-RoBERTa show impressive cross lingual transfer of their capabilities from English to other languages. In this work, we propose a pretraining strategy that uses active forgetting to achieve similar cross lingual transfer in decoder-only LLMs. We show that LLMs pretrained with active forgetting are highly effective when adapting to new and unseen languages. Through extensive experimentation, we find that LLMs pretrained with active forgetting are able to learn better multilingual representations which translates to better performance in many downstream tasks.

1 Introduction

Despite demonstrating excellent performance on English, LLM performance on multilingual benchmarks is often limited (Ahuja et al., 2023, 2024). A common method to introduce new languages to existing LLMs involves vocabulary expansion and retraining token embeddings (Balachandran, 2023; Cui et al., 2024). In many cases, these models are further finetuned on translations of English instruction tuning datasets such as Li et al. (2023); Wei et al. (2023); Singh et al. (2024). Building multilingual LLMs by simply having a large number of languages in the pretraining also does not work well due to the so-called "curse of multilinguality" (Conneau et al., 2020). Interestingly, encoder-only LLMs demonstrate cross lingual transfer – a phenomenon where the LLMs improve performance on non-English languages despite being trained only on English data. Past work has worked on improving such cross-lingual transfer (Pfeiffer et al., 2020; Parović et al., 2023; Chen et al., 2024) in encoder-only LLMs such as BERT or XLM-RoBERTa but less attention has been paid to cross-lingual transfer of decoder-only autoregressive LLMs.

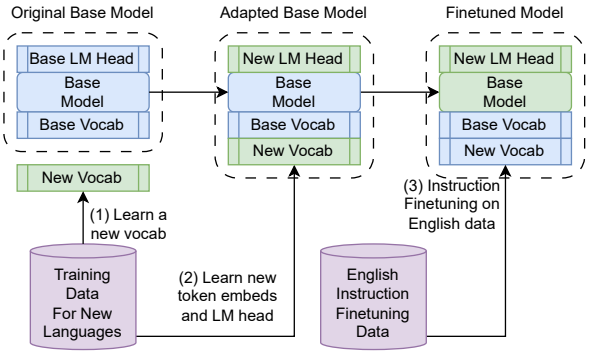


Figure 1: **Adapting base LLMs to new languages.** We show that if the base LLM is pretrained using active forgetting, it improves cross lingual transfer capabilities of the resultant English-only instruction finetuned model.

We find (Sec. 5) that the common method (depicted in Figure 1) to adapt base LLMs to a narrow set of target languages via vocabulary expansion results in improvements only to the target languages. Our results show that such adaptation significantly worsens the overall multilingual capabilities of the resultant model thereby limiting the cross lingual transfer capabilities of LLMs. Methods like zero-shot tokenizer transfer (Minixhofer et al., 2024) show promising results when adapting (decoder-only) base LLMs to new tasks such as programming *without* vocabulary expansion, but their applicability in the multilingual context has not been studied well. On the other hand, (Chen et al., 2024) show that training encoder-only LLMs with active forgetting leads to better language "plasticity" and further improves cross lingual transfer.

In this work, we want to improve the cross lingual transfer abilities of autoregressive LLMs by pretraining them using active forgetting, proposed

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by Chen et al. (2024). We show that models pretrained via active forgetting are better at adapting to new languages with lower degradation in performance on other languages. We also show that models pretrained with active forgetting have better perplexity and isotropy as compared to vanilla pretrained and adapted large language models. Our contributions are listed below:

1. We find that the common method of adapting base LLMs to newer languages (Figure 1) leads to improvements in performance *only* on the newer languages at the cost of performance of other languages.

2. We illustrate that base LLMs pretrained with active forgetting lead to higher quality multilingual representations.

3. These improved representations also lead to better cross lingual transfer. Active forgetting based LLMs outperform the baselines on 6 out of 7 multilingual benchmarks.

2 Related Work

Multilingual Language Modelling Efforts like BLOOM (BigScience Workshop et al., 2023) and PolyLM (Wei et al., 2023) have created large multilingual LLMs, however, instruction finetuned LLMs are more desirable over pretrained models in real world usecases due to their zero shot instruction following capabilities. Multilingual instruction tuning datasets have been introduced by Wei et al. (2023), Li et al. (2023) and Singh et al. (2024) to further enhance the instruction following capabilities of these models in multilingual settings. Most of these datasets are synthetically generated except Aya which is human curated. The costs of high quality human annotations show the need for models with high cross lingual transfer capabilities.

Cross Lingual Transfer Earlier works have shown that crosslingual transfer can be beneficial for multilingual pretrained models to gain task ability from English-only labelled data (Rajaee and Monz, 2024; Deb et al., 2023; Parović et al., 2023; Zhao et al., 2024b). While these techniques are effective to a certain extent, they are limited by the multilingual abilities of the pretrained model. Moreover, they cannot be effective for the low resource settings even on multilingual models due to poorer representation of rarer tokens in pretraining corpus and high token fertility for morphologically richer languages.

Language Adaptation Creating language models that can learn newer languages successively without further pretraining or by training minimal additional parameters from unlabelled data is of significant interest to reearch community (Chen et al., 2024; Pfeiffer et al., 2020; Zhao et al., 2024a). While these techniques improve the language capabilities on the newer languages, our results show that this improvement comes at the cost of performance on other languages. We focus on improving the pretraining of the base LLM itself with the intention of improving performance of such language adaptation techniques post training.

3 Method

Chern et al. (2023) propose using "active forgetting" based pretraining where token embeddings of the model are reset to random embeddings after every k steps of pretraining. They find that using active forgetting to pretrain encoder-only models improves their cross lingual transfer i.e. finetuning only on task-specific English labelled data improves task performance on non-English languages as well. In this work, we study benefits of active forgetting to train and adapt decoder-only models to new and unseen languages. Figure 1 shows standard procedure of introducing new languages to the base LLM through vocabulary expansion.

Specifically, we are given a base LLM $\mathcal{M}_{\text{base}}$ with vocabulary $\mathcal{V}$ which we wish to adapt to L new languages. We assume access to a reasonably sized corpus $\mathcal{D}^{L}_{\text{train}}$ consisting of unstructured text of L languages. In the adaptation process, first a new vocabulary $\mathcal{V}^{L}$ is learned over $\mathcal{D}^{L}_{\text{train}}$ and merged with $\mathcal{V}$ to form a larger vocabulary $\mathcal{V}_{\text{merged}}$. In the second stage, the language modeling head of $\mathcal{M}$ is replaced to be of the appropriate size i.e. $|\mathcal{V}_{\text{merged}}|$. Then the new language modeling head and token embeddings of newly added tokens (i.e. $\mathcal{V}_{\text{merged}} - \mathcal{V}$) are learned with standard language modeling training over $\mathcal{D}^{L}_{\text{train}}$. Notice that the entire Transformer stack of $\mathcal{M}$ and token embeddings of $\mathcal{V}$ are held frozen during this training. The resultant model at the end of second stage is denoted by $\mathcal{M}_{\text{adapted}}$ and has language modeling head of the size $|\mathcal{V}_{\text{merged}}|$. In the final stage, we instruction finetune the $\mathcal{M}_{\text{adapted}}$ on English only data to get $\mathcal{M}^{\text{finetuned}}_{\text{adapted}}$ which is evaluated on multilingual benchmarks to assess its cross lingual transfer capoabilities. Our hypothesis is that if $\mathcal{M}_{\text{base}}$ is pretrained using active forgetting, the corresponding

$\mathcal{M}_{\text{adapted}}^{\text{finetuned}}$ will be better at cross lingual transfer.

4 Experiments

Training Setup We pretrain our $\mathcal{M}_{\text{base}}$ on Wikipedia dumps[1] of 12 languages (referred as "pretraining" languages) from Shaham et al. (2024). The adaptation dataset $\mathcal{D}_{\text{train}}^{L}$ consists of Wikipedia dumps of 14 new languages (referred as "adapting" languages) disjoint from the pretraining languages. The exact languages are presented in Table 6. In our results, "BA" refers to "Baseline Adapted" i.e. $\mathcal{M}_{\text{adapted}}^{\text{finetuned}}$ where $\mathcal{M}_{\text{base}}$ was trained with standard optimization. "AFA" refers to "Active Forgetting Adapted" i.e. $\mathcal{M}_{\text{adapted}}^{\text{finetuned}}$ where $\mathcal{M}_{\text{base}}$ was trained with active forgetting. We also present results on "Baseline" which refers to $\mathcal{M}_{\text{base}}^{\text{finetuned}}$ i.e. instruction tuned $\mathcal{M}_{\text{base}}$ without adaptation. We experiment with $\mathcal{M}_{\text{base}}$ of 3 different sizes and use OpenOrca (Lian et al., 2023) as our English instruction tuning dataset and contains 2.91M data points.

Evaluation Setup We follow Aggarwal et al. (2024) to evaluate multilingual capabilities of our models using 6 multilingual benchmarks. Additionally, we establish superiority of the active forgetting pretrained models by measuring isotropy of their embeddings (Ethayarajh, 2019) and model perplexity on 50 languages (26 new languages not in "pretraining" or "adapting" as shown in Table 6) in mC4 (Xue et al., 2021). We also evaluate the 4-shot translation (English-to-X) performance of the models to the same set of 50 languages using the FLORES-200 dataset (NLLB Team et al., 2022).

5 Discussion

Active Forgetting Leads to Better Language Adaptation We study the intrinsic properties of the adapted models $\mathcal{M}_{\text{adapted}}$ using perplexity on mC4 and isotropy i.e. self similarity of contextual embeddings (Ethayarajh, 2019). As shown in Table 1, we find that AFA models achieve consistently lower perplexity than both "Baseline" and "BA". Moreover, AFA models are also able to better contextualize a sentence over all languages as observed by lower self similarity (isotropy) scores in Table 2. This suggests that the quality of multilingual representations of AFA is better than other models.

Active Forgetting Improves Cross-Lingual Transfer In Figure 12, we compare performance of our models on various multilingual benchmarks similar to Aggarwal et al. (2024) and the translation task. We find that despite instruction tuning only on English, AFA models show improvements across all language classes. AFA outperforms both Baseline and BA models on 6 out of 7 tasks in our evaluation suite. More importantly, we find

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	**25.041**	31.440	34.663	31.451
BA	25.097	31.405	36.993	32.573
AFA	25.180	**30.373**	**34.345**	**31.033**
Number of parameters = 782M				
Baseline	22.826	29.382	31.099	28.633
BA	23.155	28.924	33.766	29.864
AFA	**22.727**	**27.949**	**31.047**	**28.183**
Number of parameters = 1.6B				
Baseline	20.745	26.831	28.497	26.170
BA	20.828	26.117	30.616	27.007
AFA	**20.654**	**25.048**	**28.386**	**25.596**
Number of parameters = 2.8B				
Baseline	20.887	26.345	28.689	26.198
BA	20.958	25.969	30.768	27.034
AFA	**20.716**	**24.858**	**28.395**	**25.621**

Table 1: Detailed results on perplexity (Lower is Better). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages.

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	0.683	0.663	0.667	0.670
BA	0.659	0.651	0.678	0.666
AFA	**0.640**	**0.624**	**0.640**	**0.636**
Number of parameters = 782M				
Baseline	0.610	0.607	0.612	0.610
BA	0.602	0.593	0.618	0.607
AFA	**0.587**	**0.566**	**0.588**	**0.582**
Number of parameters = 1.6B				
Baseline	0.549	0.550	0.562	0.555
BA	0.555	0.548	0.560	0.555
AFA	**0.531**	**0.513**	**0.530**	**0.525**
Number of parameters = 2.8B				
Baseline	**0.504**	**0.506**	**0.506**	**0.505**
BA	0.506	0.508	**0.506**	0.507
AFA	**0.504**	**0.506**	**0.506**	**0.505**

Table 2: Detailed results on isotropy (Lower is Better). All the abbrevations are same as in Table 1.

[1] https://huggingface.co/datasets/wikimedia/wikipedia 20231101 dump

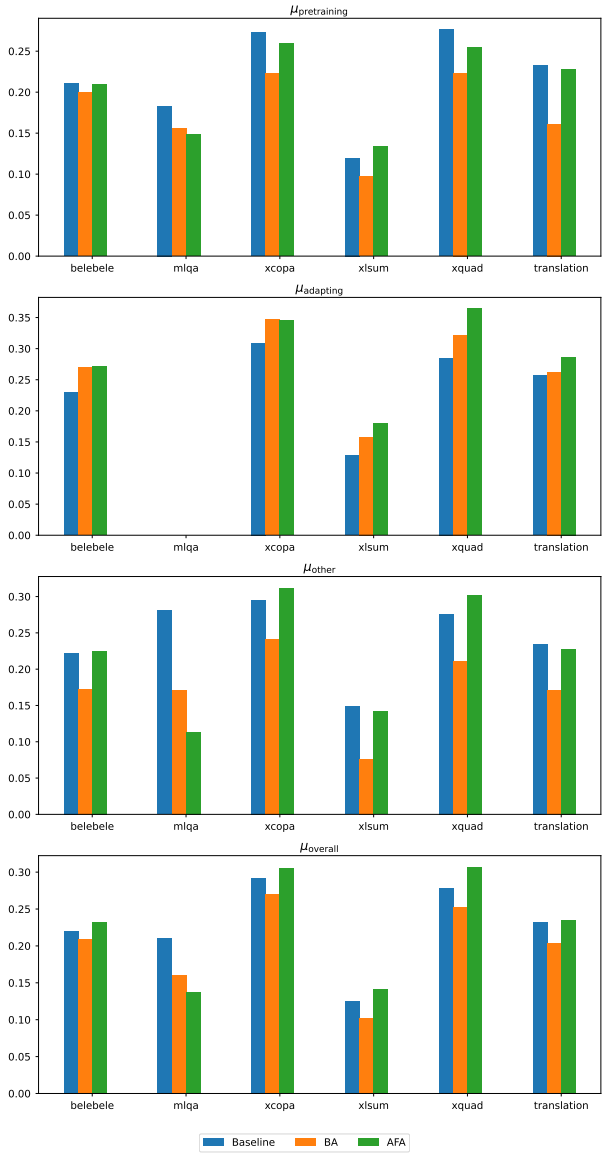


Figure 2: Task wise performance comparison for the 2.8 billion parameter models. Higher is better for all tasks.

Model	$\mu_{pretraining}$	$\mu_{adapting}$	μ_{other}	$\mu_{overall}$
Number of parameters = 400M				
Baseline	0.080	0.084	**0.100**	0.091
BA	**0.101**	0.092	0.074	0.086
AFA	0.078	**0.103**	0.098	**0.094**
Number of parameters = 782M				
Baseline	**0.162**	0.138	0.154	0.152
BA	0.127	**0.208**	0.119	0.146
AFA	0.158	0.190	**0.180**	**0.178**
Number of parameters = 1.6B				
Baseline	**0.208**	0.197	**0.221**	0.211
BA	0.134	**0.274**	0.147	0.180
AFA	0.202	0.254	0.198	**0.215**
Number of parameters = 2.8B				
Baseline	**0.241**	0.255	**0.245**	0.237
BA	0.163	0.255	0.174	0.205
AFA	0.240	**0.288**	0.229	**0.239**

Table 3: Detailed results on translation (en-to-XX) on the subset of languages from FLORES-200 (NLLB Team et al., 2022). All the abbrevations are same as in Table 1 (Higher is better). We use BLEU Score as the metric (Papineni et al., 2002).

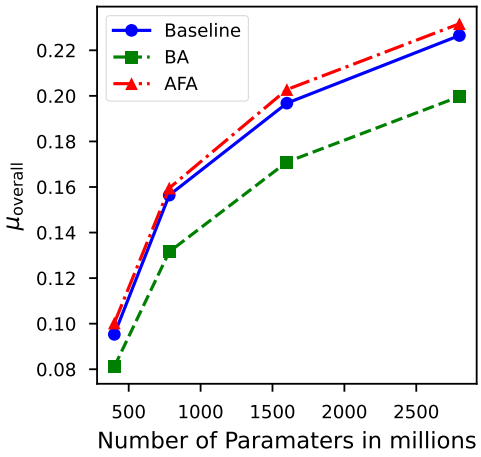


Figure 3: **Effect of model scale.** Average task performance of models against the model parameter count. We consider the same tasks in figure 2

that BA models often are worse overall ($\mu_{overall}$) as compared to Baseline. This reaffirms findings by Shaham et al. (2024) where if the base model is already multilingual, adapting to a narrow set of languages can *worsen* the overall performance. AFA models on the other hand do not seem to suffer from the same limitation.

Analysis on Language Class and Model Size We study how language adaptation affects performance on each language class ("pretraining", "adapting" or "other") by studying performance on translation in Table 3. We find that BA models show significantly better performance (as compared to Baseline) on "adapting" languages at all model scales. Moreover, improvement of BA over Baseline is larger than improvement of AFA over Baseline for larger models and BA models seem to degrade performance on all other language classes leading to worse overall($\mu_{overall}$) scores as observed in Figure 3.

6 Conclusion and Future Work

In this work, we show that pretraining with active forgetting can improve language adaptability and cross lingual transfer capabilities of autoregressive (decoder-only) language models. In our experi-

ments, we found that a base LLM that is pretrained with active forgetting and instruction tuned only with English data leads to improvements across all languages in 6 out of 7 multilingual benchmarks. We observed this behavior to be consistent at all the model sizes we tried. The improvements in these downstream tasks can be attributed to the active forgetting models learning better multilingual representations. We hope that these findings encourage pretraining of larger LLMs with active forgetting. Future work can also explored effective language adaptation methods to adapt an *existing* finetuned LLM to new languages.

Limitations

The methods described in this work are aimed at improving *pretraining* of multilingual LLMs. As such, these cannot be directly applied to existing LLMs. An interesting direction to explore could be to take intermediate checkpoints of open source LLMs such as TinyLlama (Zhang et al., 2024) or OLMo (Groeneveld et al., 2024) and simulate active forgetting by resetting their embeddings then continuing to train on $\mathcal{D}^L_{\text{train}}$. Finally, our training and evaluation suite consisted primarily of training language models of size that could comfortably fit in our compute budget. Moreover, the data is used (10 billion tokens in total) is much lesser than models like XLM-R which were trained on much larger data with more than 100 billion in much more languages, which can give state of the art performance on our evaluation with much lesser model parameter size, since the data is larger, the compute FLoPs used to train XLM-R is much larger than what we used despite our models being larger. Further experiments and evaluation is needed to study efficacy of pretraining with active forgetting on larger scale models.

Ethics Statement

The proposed method of pretraining directly affects the token embeddings of an LLM. While we find that these lead to better representations in terms of multilinguality, special care must be taken before deploying such LLMs. A thorough study of their overall capabilities as well as intrinsic and extrinsic biases must be performed before deploying such LLMs to any public facing interface.

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Hyperparameter	Value
Learning rate	1×10^{-4}
Number of steps	150,000
Global batch size	128
Block size	4096
Scheduler	Cosine
Warmup	Linear
Warmup steps	10%
Optimizer	AdamW (Loshchilov and Hutter, 2019)
Weight decay	0
Embed. reset steps	10,000

Table 4: Hyperparameters for pretraining. Only the active forgetting models reset their token embeddings every "Embed. reset steps". Note that embeddings are *not* reset after the final step.

Hyperparameter	Value
Learning rate	1×10^{-6}
Epochs	5
Global batch size	16
Scheduler	Cosine
Warmup	Linear
Warmup steps	10%
Optimizer	AdamW (Loshchilov and Hutter, 2019)
Weight decay	0

Table 5: Hyperparameters for finetuning.

A Details on Computational Resources

Our base LLM uses Mistral architecture downscaled to fit our compute resources. We explore 3 configurations with total parameter counts of 400M, 782M and 1.6B respectively by reducing the hidden dimensions, number of attention heads and the total number of Transformer blocks. The vocabulary size of the base model was kept fixedd at 32000 while the adapting vocabulary was allowed to merge 16000 more tokens leading to $|\mathcal{V}_{\text{merged}}| = 48000$. All our experiments are run on a single NVIDIA A100 GPU with 80 GB VRAM. The total GPU hours for all experiments and evaluations come out to roughly 650 hours. The hyperparameters for pretraining and finetuning runs are presented in Table 4 and Table 5 respectively.

B Details on Language Classes

In Table 6, we present the 2 main classes of languages relevant to our experiments. Specifically, all base models (with or without active forgetting) are pretrained on the "pretraining" languages. Each base model is then adapted to the 14 "adapting" languages together. Any language that is not in either of these 2 classes is considered "other". The "other" in Table 6 refers to the additional languages we used for perplexity and isotropy analysis.

group	languages
pretraining (12)	ar, zh, cs, en, et, fi, he, hi, it, ru, es, sw
adapting (14)	ja, fr, pt, nl, se, tr, da, no, ko, pl, hu, th, mr, gu
other (26)	af, bg, bn, de, id, ml, sv, ta, te, ur, vi, tg, ka, sq, ps, sr, az, my, co, iw, mn, st, sk, ha

Table 6: Language Details

C Evaluation Setting and Prompts

We use lm-evaluation-harness [2] for our evaluation experiments with default settings.

The evaluation prompts for all the tasks in our evaluation suite (6 tasks from Aggarwal et al. (2024) and Translation on FLORES-200) are presented in Figure **??** to Figure 9.

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to perform open-domain
commonsense causal reasoning. You will
be provided a premise and two alternatives,
where the task is to select the alternative
that more plausibly has a causal relation
with the premise. Answer as concisely as
possible in the same format as the examples
below:

Given this premise:
{{premise}}
What's the best option?
-choice1 : {{choice1}}
-choice2 : {{choice2}}
We are looking for {% if question == "cause"
%} a cause {% else %} an effect {% endif %}
<|im_end|>
<|im_start|>assistant
```

Figure 4: XCOPA Prompt

[2] https://github.com/EleutherAI/lm-evaluation-harness

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to perform reading comprehension
task. Given the following passage, query,
and answer choices, output the letter
corresponding to the correct answer.

Passage: {{flores_passage}}
Query: {{question}}
Choices:
A: {{mc_answer1}}
B: {{mc_answer2}}
C: {{mc_answer3}}
D: {{mc_answer4}}

<|im_end|>
<|im_start|>assistant
```

Figure 5: Belebele Prompt

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to solve reading comprehension
problems. You will be provided questions
on a set of passages and you will need
to provide the answer as it appears in
the passage. The answer should be in the
same language as the question and the passage.

Context:{{context}}
Question:{{question}}

Referring to the passage above, the
correct answer to the given question is

<|im_end|>
<|im_start|>assistant
```

Figure 6: MLQA Prompt

D Detailed Results on All Model Scales and Tasks

Table 7 to Table 11 and Figure 10 and Figure 11 present detailed results on all tasks at all model scales and language classes.

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	**0.117**	0.113	0.095	0.107
BA	0.075	0.108	0.092	0.091
AFA	0.090	**0.133**	**0.110**	**0.110**
Number of parameters = 782M				
Baseline	**0.160**	0.158	0.157	0.158
BA	0.119	0.188	0.138	0.146
AFA	0.148	**0.211**	**0.166**	**0.173**
Number of parameters = 1.6B				
Baseline	**0.184**	0.199	0.198	0.194
BA	0.167	0.236	0.156	0.182
AFA	**0.184**	**0.241**	**0.206**	**0.209**
Number of parameters = 2.8B				
Baseline	0.208	0.224	0.225	0.219
BA	0.195	0.261	0.183	0.208
AFA	**0.215**	**0.279**	**0.227**	**0.237**

Table 7: Detailed results on belebele (Bandarkar et al., 2023). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages. We use Accuracy as the metric.

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	0.099	N/A	0.048	0.085
BA	0.059	N/A	0.069	0.062
AFA	**0.106**	N/A	**0.097**	**0.104**
Number of parameters = 782M				
Baseline	**0.141**	N/A	**0.130**	**0.138**
BA	0.112	N/A	0.100	0.108
AFA	0.099	N/A	0.093	0.098
Number of parameters = 1.6B				
Baseline	**0.151**	N/A	**0.248**	**0.179**
BA	0.139	N/A	0.149	0.141
AFA	0.135	N/A	0.103	0.126
Number of parameters = 2.8B				
Baseline	**0.183**	N/A	**0.276**	**0.210**
BA	0.171	N/A	0.171	0.171
AFA	0.167	N/A	0.126	0.156

Table 8: Detailed results on mlqa (Lewis et al., 2020). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages. We use F1-abstractive score as the metric.

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	**0.131**	0.111	0.107	0.116
BA	0.086	0.149	0.088	0.098
AFA	0.112	**0.164**	**0.110**	**0.121**
Number of parameters = 782M				
Baseline	**0.227**	0.204	0.204	0.212
BA	0.161	0.222	0.154	0.169
AFA	0.219	**0.245**	**0.216**	**0.223**
Number of parameters = 1.6B				
Baseline	0.238	0.275	**0.263**	0.256
BA	0.200	0.319	0.210	0.226
AFA	**0.254**	**0.321**	**0.263**	**0.270**
Number of parameters = 2.8B				
Baseline	0.278	0.319	0.288	0.295
BA	0.243	**0.350**	0.253	0.282
AFA	**0.279**	0.346	**0.299**	**0.308**

Table 9: Detailed results on xcopa (Ponti et al., 2020). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages. We use accuracy as the metric.

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to solve reading comprehension
problems. You will be provided questions
on a set of passages and you will need
to provide the answer as it appears in
the passage. The answer should be in the
same language as the question and the passage.

Context:{{context}}
Question:{{question}}

Referring to the passage above, the
correct answer to the given question is

<|im_end|>
<|im_start|>assistant
```

Figure 7: XQUAD Prompt

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	**0.060**	0.031	0.037	0.052
BA	0.039	0.024	**0.054**	0.039
AFA	0.056	**0.095**	0.048	**0.060**
Number of parameters = 782M				
Baseline	**0.085**	0.074	0.084	0.083
BA	0.055	0.100	0.052	0.061
AFA	0.076	**0.113**	**0.095**	**0.084**
Number of parameters = 1.6B				
Baseline	0.096	0.119	**0.110**	0.101
BA	0.076	0.115	0.058	0.079
AFA	**0.097**	**0.134**	0.109	**0.104**
Number of parameters = 2.8B				
Baseline	**0.122**	0.141	**0.138**	**0.127**
BA	0.101	0.119	0.082	0.100
AFA	0.110	**0.162**	0.114	0.124

Table 10: Detailed results on xlsum (Hasan et al., 2021). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages. We use Rouge Score as the metric.

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to summarize any given
article. You should summarize all important
information concisely in the same language
in which you have been provided the document.
Following the examples provided below:

{{text}}

<|im_end|>
<|im_start|>assistant
```

Figure 8: XLSUM Prompt

```
<|im_start|>system
You are a large language model trained to
solve multiple NLP tasks accurately. For any
given NLP task, you must produce an output
that is factually correct and succinct.
<|im_end|>
<|im_start|>user

The task is to translate the given sentence
in English to language {{language}}. There
are 4 examples provided below. Produce the
translation of the 5th sentence:

{{text}}

<|im_end|>
<|im_start|>assistant
```

Figure 9: Translation Prompt

Model	$\mu_{\text{pretraining}}$	μ_{adapting}	μ_{other}	μ_{overall}
Number of parameters = 400M				
Baseline	**0.138**	0.114	0.098	**0.121**
BA	0.093	0.178	**0.104**	0.111
AFA	0.096	**0.233**	0.077	0.112
Number of parameters = 782M				
Baseline	0.186	0.185	**0.213**	0.195
BA	0.147	**0.225**	0.145	0.159
AFA	**0.191**	0.218	0.211	**0.202**
Number of parameters = 1.6B				
Baseline	**0.243**	0.248	0.231	0.240
BA	0.195	0.285	0.216	0.217
AFA	0.242	**0.335**	**0.258**	**0.263**
Number of parameters = 2.8B				
Baseline	0.258	0.277	0.254	0.263
BA	0.230	0.320	0.230	0.260
AFA	**0.265**	**0.361**	**0.230**	**0.308**

Table 11: Detailed results on xquad (Artetxe et al., 2020). BA refers to Baseline adapted model. AFA refers to Active Forgetting adapted model. $\mu_{\text{pretraining}}$ refers to performance averaged over languages in the pretraining split. μ_{adapting} refers to averaging over languages in the adapting split. μ_{other} refers to averaging on languages that are in neither split. μ_{overall} refers to the average over all languages. We use F1 abstractive as the metric.

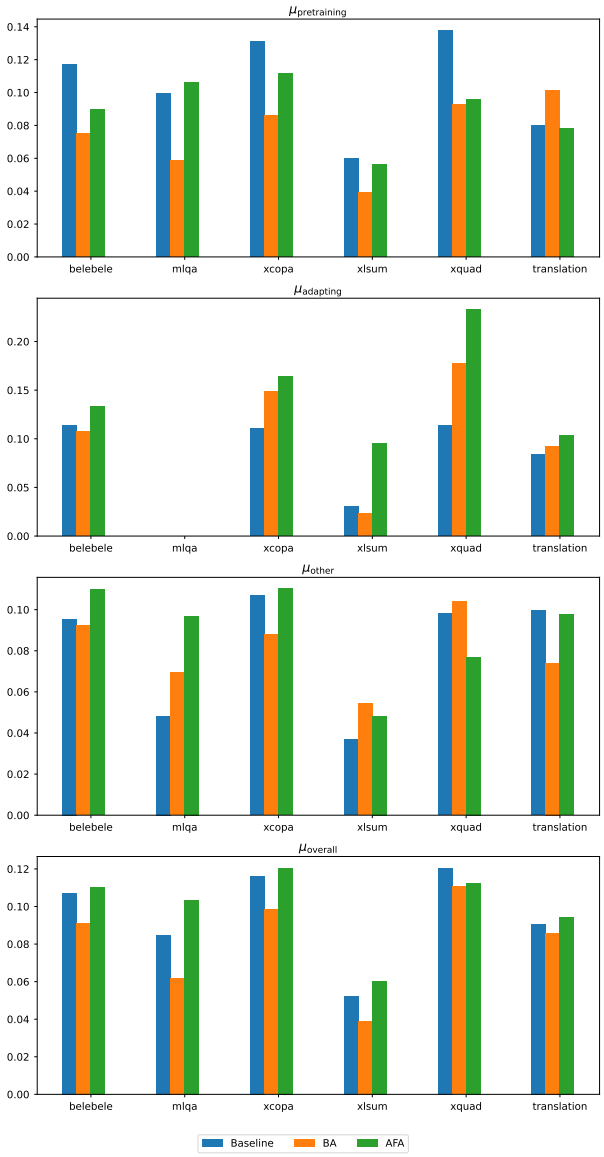


Figure 10: **Task wise performance comparisonfor 782 million parameter models.** We find that the "Baseline Adaptation" method is able to improve performance only on adapting languages, often at the cost of performance on all other languages.

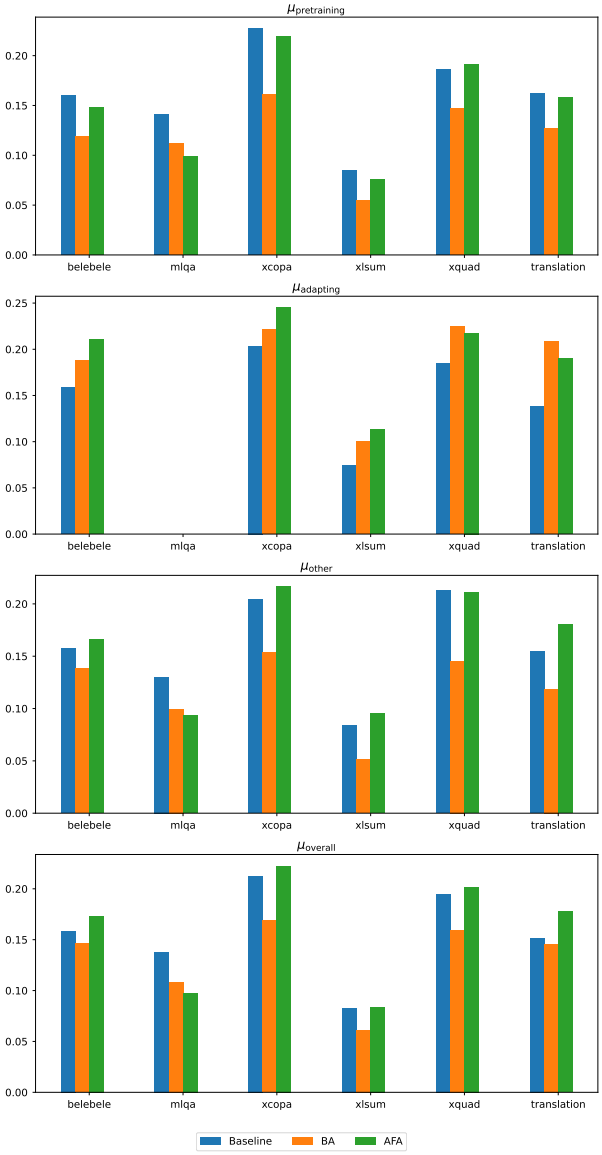


Figure 11: **Task wise performance comparison for 400 million parameter models.** We find that the "Baseline Adaptation" method is able to improve performance only on adapting languages, often at the cost of performance on all other languages.

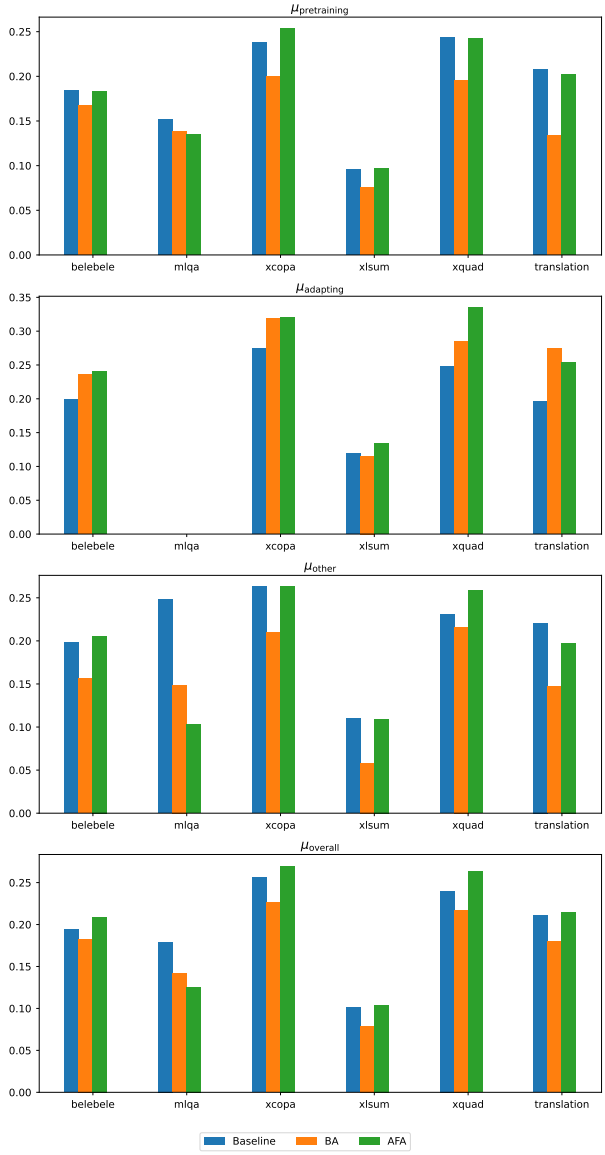


Figure 12: **Task wise performance comparison for the 1.6 billion parameter models.** We find that the "Baseline Adaptation" method is able to improve performance only on adapting languages, often at the cost of performance on all other languages.